THE HOM-TENSOR ADJUNCTION FOR DIFFERENTIAL GRADED MODULES OVER A DIFFERENTIAL GRADED ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the common bilinear description, the translation through more-algebra-lemma-compose and the commutativity constraint, and the explicit two-expansion sign computation for the A-bilinearity closed by the parity check are all retained verbatim (the author marks the final parity check as a trivial mod-2 calculation; the exponents are displayed in full).

Editorial note. The label lemma-characterize-hom-other-side names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Bilinear description of morphisms into a Hom complex. Same chapter, Section section-tensor-product. Fixes the right-hand side: maps of complexes from the tensor product correspond to graded bilinear maps compatible with differentials, the target description of the adjunction.
- Maps out of a total tensor product. More on Algebra, Lemma more-algebra-lemma-compose. Determines, from an element of the left-hand side, the map of complexes out of the total tensor product, the starting point of the translation.

Source reference. Original lemma 0A5Y: sources/stacks-chow/tags.txt, source lines 12826–12826. Prerequisite pointer invoked at proof line(s) 1906; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00CV: sources/stacks-dga/pinned-original-source.tex, source lines 1606–1606. Prerequisite pointer invoked at proof line(s) 1895; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let (A, d) be a differential graded R-algebra. Let M be a right differential graded A-module and let M' be a left differential graded A-module. Let $N^\bullet$ be a complex of R-modules. Then we have*

$$\mathrm{Hom}_{\textit{left diff graded } A\textit{-modules}}(M', \mathrm{Hom}(M, N^\bullet)) = \mathrm{Hom}_{Comp(R)}(M \otimes_A M', N^\bullet)$$

where $M \otimes_A M'$ is viewed as a complex of R-modules as in Section 00CV.

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Proof. Let us show that both sides correspond to graded A-bilinear maps

$$M \times M' \longrightarrow N^\bullet$$

compatible with differentials. We have seen this is true for the right hand side in Section 00CV. Given an element g of the left hand side, the equality of More on Algebra, Lemma 0A5Y determines a map of complexes $g' : \mathrm{Tot}(M' \otimes_R M) \to N^\bullet$. We precompose with the commutativity constraint to get

$$\mathrm{Tot}(M \otimes_R M') \xrightarrow{\psi} \mathrm{Tot}(M' \otimes_R M) \xrightarrow{g'} N^\bullet$$

which corresponds to a graded R-bilinear map $g'' : M \times M' \to N^\bullet$ compatible with differentials. The A-linearity of g translates immediately into A-bilinarity of g''. Namely, say $x \in M^e$ and $x' \in (M')^{e'}$ and $a \in A^n$. Then on the one hand we have

$$\begin{aligned} g''(x, ax') &= (-1)^{e(n+e')} g'(ax' \otimes x) \\ &= (-1)^{e(n+e')} g(ax')(x) \\ &= (-1)^{e(n+e')} (a \cdot g(x'))(x) \\ &= (-1)^{e(n+e')+n(n+e+e')+n} g(x')(xa) \end{aligned}$$

and on the other hand we have

$$g''(xa, x') = (-1)^{(e+n)e'} g'(x' \otimes xa) = (-1)^{(e+n)e'} g(x')(xa)$$

which is the same thing by a trivial mod 2 calculation of the exponents. □